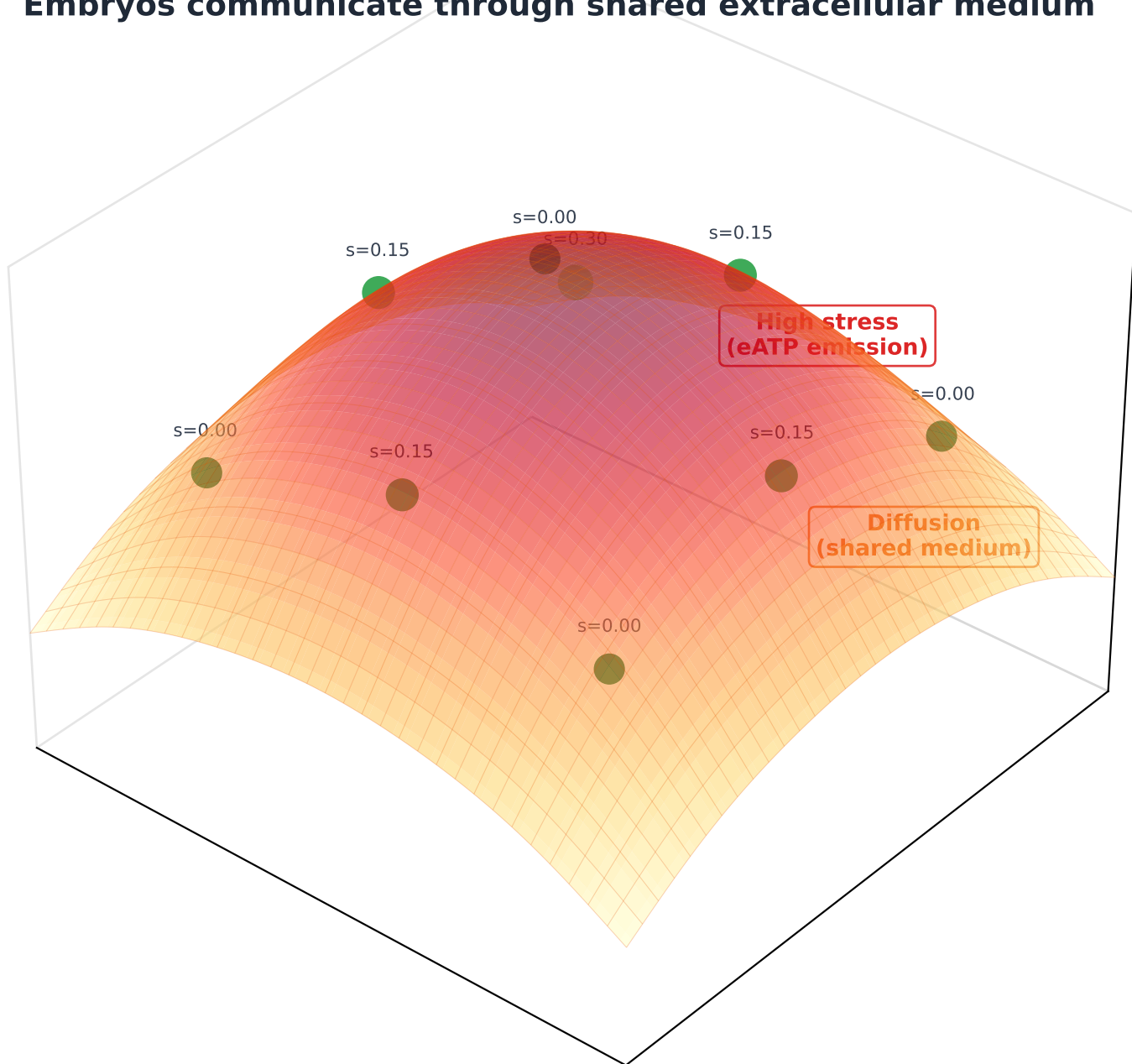


MULTI-EMBRYO STRESS RESCUE NETWORK

Bioelectric fields within embryos generate stress signals that diffuse between embryos

COLLECTIVE STRESS FIELD

Embryos communicate through shared extracellular medium



Embryo Stress Level

- Healthy ($s = 0.0$)
- Mildly stressed ($s = 0.15$)
- Strongly stressed ($s = 0.30$)

Stress Field Equation

$$\frac{dF}{dt} = D_F \nabla^2 F - \gamma_F F + \text{emission}$$

$\text{emission} = \text{mean}(S) \text{ per embryo}$

Rescue Mechanism

Stressed embryos emit eATP into the shared medium. Collective signal reshapes each receiver's ATP dynamics via saddle-node bifurcation.

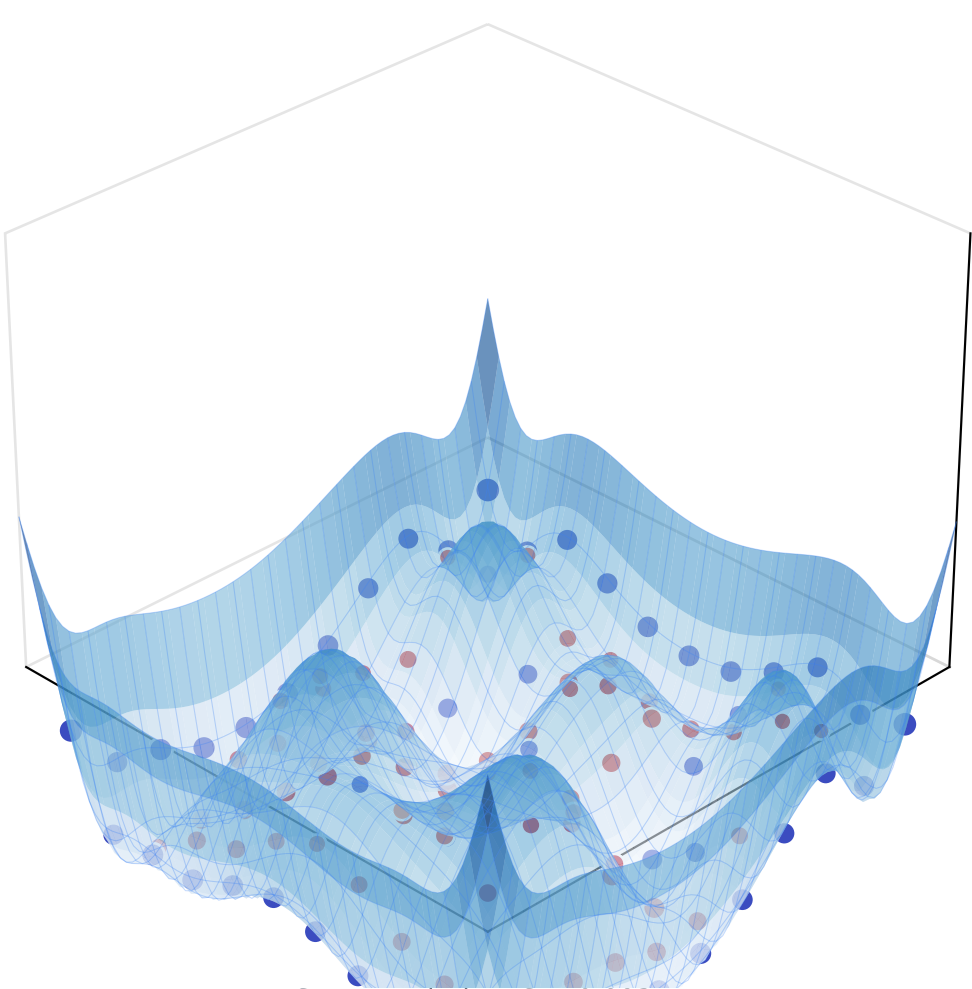
Individual cell stress (S) → Collective extracellular field (F)

INDIVIDUAL EMBRYO BIOELECTRIC FIELDS (11×11 cells)

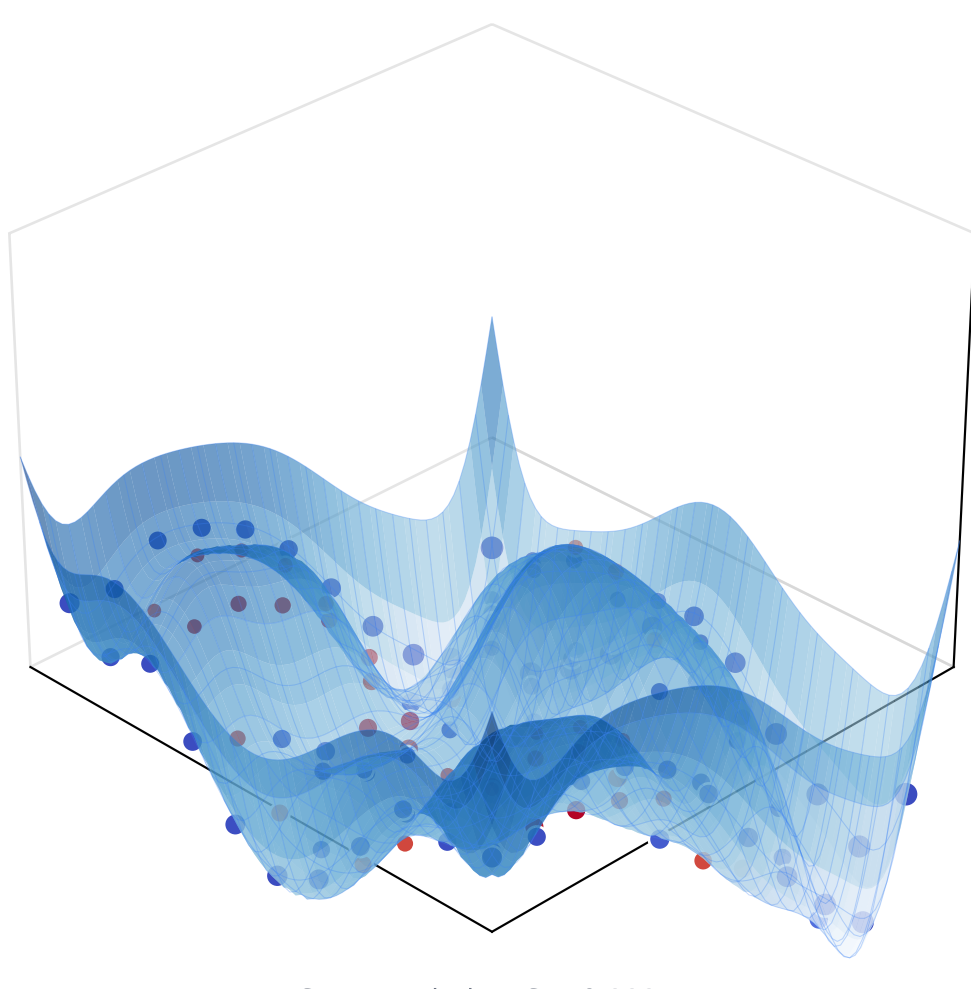
HEALTHY ($s=0.0$)
Intact bioelectric field

MILDLY STRESSED ($s=0.15$)
Partially disrupted field

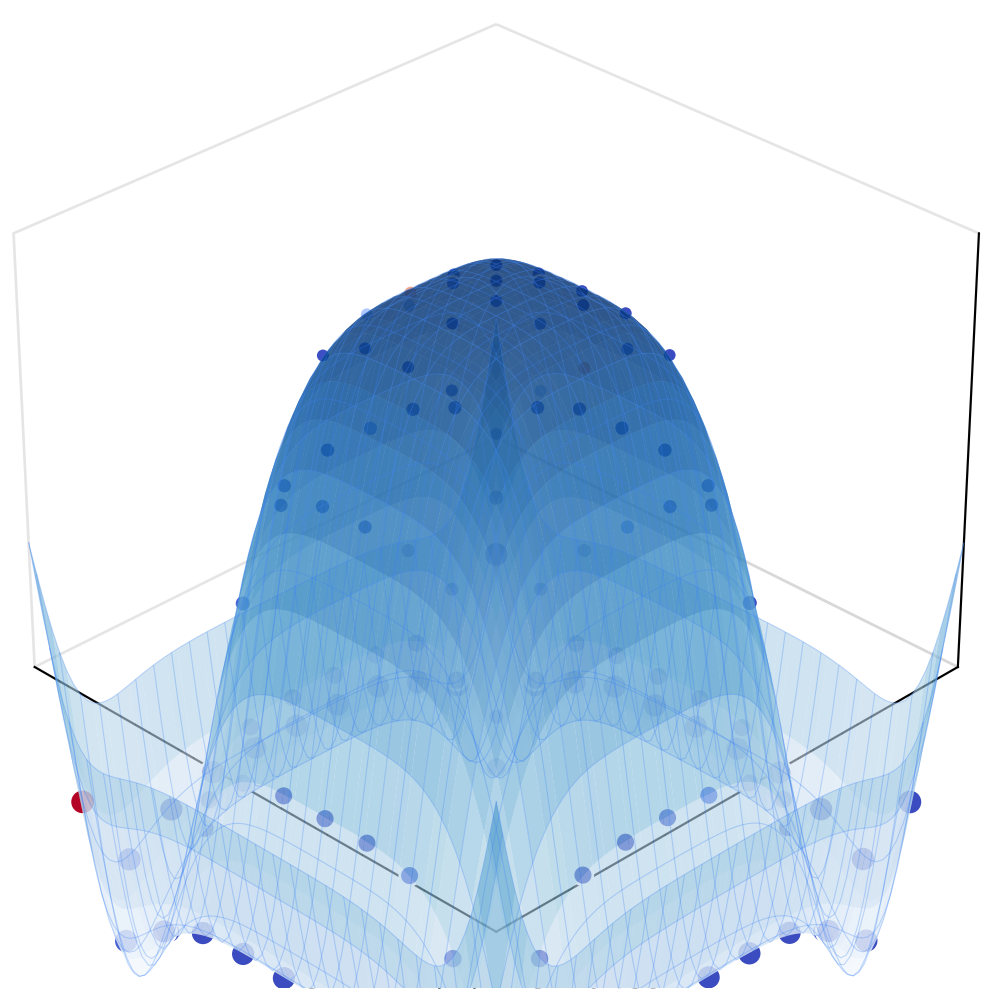
STRONGLY STRESSED ($s=0.30$)
Collapsed field pattern



Stress emission: $S = 0.093$



Stress emission: $S = 0.111$



Stress emission: $S = 0.138$